ORF 524 — Fall 2026

Week 06

Note

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} E[X_i] = \mu$$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \bar{X}_n)^2$$

$$= \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \mu)^2 + (\bar{X}_n - \mu)^2$$

Delta Method

1) $r_n(\hat{\theta}_n - \theta) \rightsquigarrow \mathcal{L}$; $r_n \to \infty$ $\quad (r_n = \sqrt{n})$

2) $g : \Theta \mapsto \mathbb{R}$, smooth (and more)

Study $\quad g(\hat{\theta}_n) - g(\theta)$

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1) $\Rightarrow \quad \hat{\theta}_n - \theta = O_p(r_n^{-1}) \quad \Rightarrow \quad \hat{\theta} \to_p \theta$

$\hat{\theta} - \theta = o_p(1)$

2) $\Rightarrow$ Taylor expansion $\quad a \to b$

$\tilde{a}$ is in between a, b

$$g(a) - g(b) = \sum_{\ell=1}^{L} \frac{1}{\ell!} g^{(\ell)}(b) \cdot (a-b)^{\ell} + \frac{1}{(L+1)!} g^{(L+1)}(\tilde{a}) (a-b)^{L+1}$$

$$g(\hat{\theta}) - g(\theta)$$

$$= g^{(1)}(\theta)(\hat{\theta}-\theta) + \frac{1}{2} g^{(2)}(\theta)(\hat{\theta}-\theta)^2 + \frac{1}{3!} g^{(3)}(\tilde{\theta})(\hat{\theta}-\theta)^3$$

$$= O_p\left(\frac{1}{n_u}\right) + O_p\left(\frac{1}{n_u^2}\right) + o_p\left(\frac{1}{n_u^2}\right)$$

Obs 1 :

$$n_u\left(g(\hat{\theta}) - g(\theta)\right) = g^{(1)}(\theta)\, n_u \cdot (\hat{\theta}-\theta)$$

$$+ \frac{1}{2} g^{(2)}(\theta)\, \underbrace{n_u(\hat{\theta}-\theta)}_{O_p(1)} \underbrace{(\hat{\theta}-\theta) n_u}_{O_p(1)} \frac{1}{n_u}$$

$$O_p\left(n_u^{-1}\right)$$

$$g(\theta) - g(\theta) \rightsquigarrow \begin{cases} \frac{1}{n} & g^{(1)}(\theta) \cdot \ell \qquad k=1 \\ \frac{1}{n^2} & \frac{1}{2} g^{(2)}(\theta)\, \ell^2 \qquad k=2 \\ \frac{1}{n^3} & \frac{1}{3!} g^{(3)}(\theta)\, \ell^3 \qquad k=3 \\ & \vdots \\ \frac{1}{n^L} & \frac{1}{L!} g^{(L)}(\theta)\, \ell^L \qquad k=L \end{cases}$$

Choices

1) $g^{(1)}(\theta) \neq 0 \Rightarrow g^{(1)}(\theta)\, \ell$

2) $\Rightarrow g^{(1)}(\theta)\, \ell + \frac{1}{n} \cdot \frac{1}{2} g^{(2)}(\theta)\, \ell^2$

We have (at least 2) choices

$$\mathbb{P}\left[g(\hat{\theta}) - g(\theta) \le t \right] - \mathbb{P}\left[\frac{1}{\sqrt{n}} g^{(1)}(\theta) Z \le t \right] \to 0$$

$$\mathbb{P}\left[g(\hat{\theta}) - g(\theta) \le t \right] - \mathbb{P}\left[\frac{1}{\sqrt{n}} g^{(1)}(\theta) Z + \frac{1}{\sqrt{n}^2} \frac{1}{2} g^{(2)}(\theta) Z^2 \le t \right] \to 0$$

How do we handle the error? A typical self-cond

$$\frac{1}{(L+1)!} g^{(L+1)}(\tilde{\theta}) \underbrace{(\hat{\theta} - \theta)^{L+1}}_{O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}^{L+1}}\right)}$$

$$\sup_{a : |a-\theta| \le \delta} |g^{(L+1)}(a)| < C$$

for some $\delta > 0$.

Idea: $\theta = \frac{\mu_x}{\mu_y}$ $\quad \hat{\theta} = \frac{\hat{\mu}_x}{\hat{\mu}_y} = \frac{\bar{x}_n}{\bar{y}_n}$

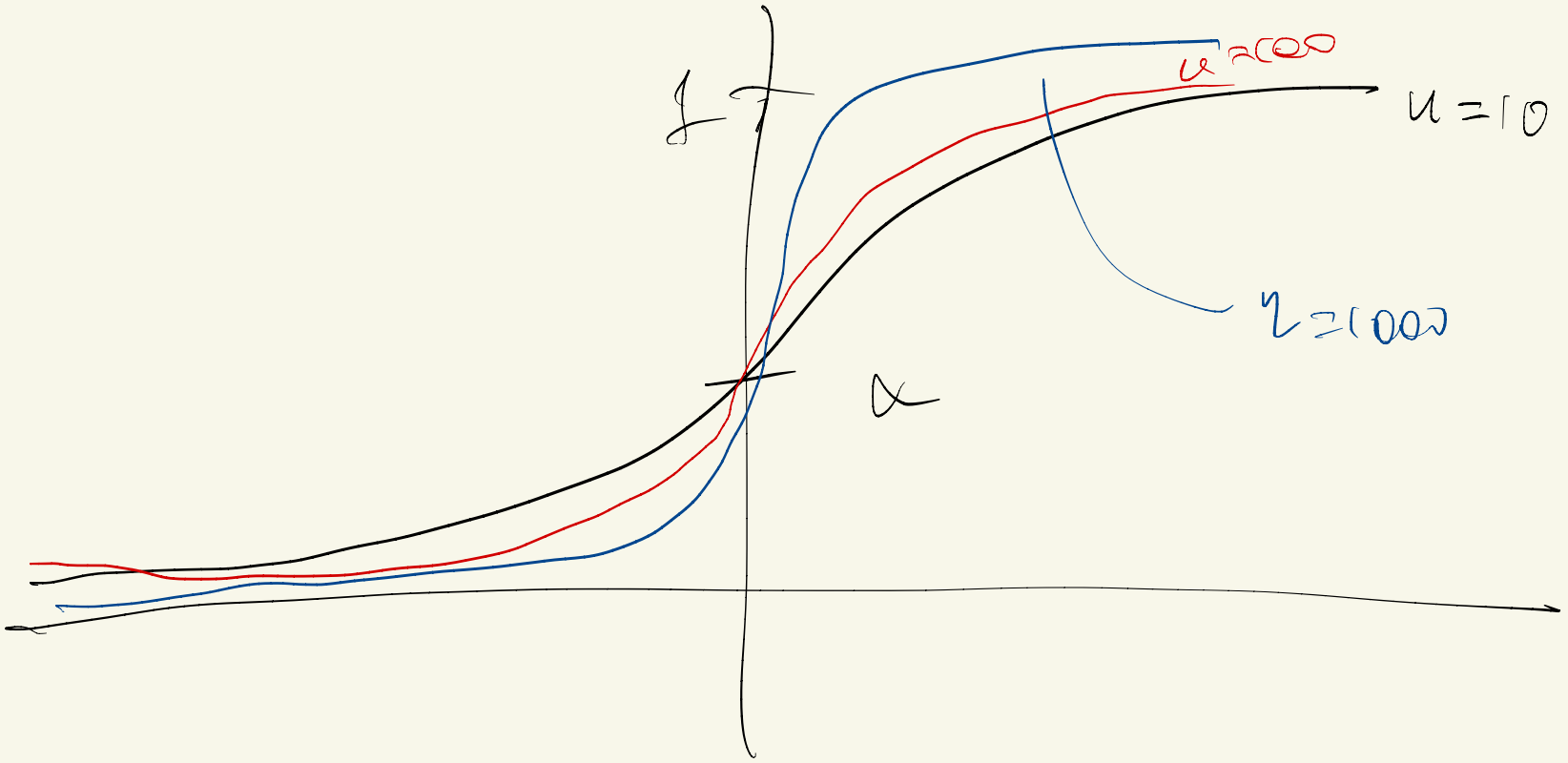
$$\sqrt{n}\left(\begin{bmatrix}\bar{x}_n \\ \bar{y}_n\end{bmatrix} - \begin{bmatrix}\mu_x \\ \mu_y\end{bmatrix}\right) \rightsquigarrow N\left(\begin{pmatrix}0 \\ 0\end{pmatrix}, V\right)$$

$$\hat{\theta} = g\begin{pmatrix}\bar{x}_n \\ \bar{y}_n\end{pmatrix} \qquad \theta = g\begin{pmatrix}\mu_x \\ \mu_y\end{pmatrix} \qquad ; g(u,v) = \frac{u}{v}$$

① Compute $\nabla g \Rightarrow \nabla g(\theta) \neq 0$

② Apply delta-method

$$\sqrt{n}\left(g\begin{pmatrix}\bar{x} \\ \bar{y}\end{pmatrix} - g\begin{pmatrix}\mu_x \\ \mu_y\end{pmatrix}\right) = \sqrt{n}(\hat{\theta} - \theta) \rightsquigarrow N(0, \nabla g(\theta)^T V \nabla g(\theta))$$


f
u=200
u=10
η=1000
x

$$y_i = x_i^T \theta + \varepsilon_i \qquad \mathbb{E}[\varepsilon_i | x_i] = 0$$

$$\dim(x_i) = d_n \to \infty \text{ as } n \to \infty$$

θ , $F_{\varepsilon|x}(e|x) = \mathbb{P}(\varepsilon_i \le e \mid x_i)$, $F_x(x) = \mathbb{P}(x_i \le x)$

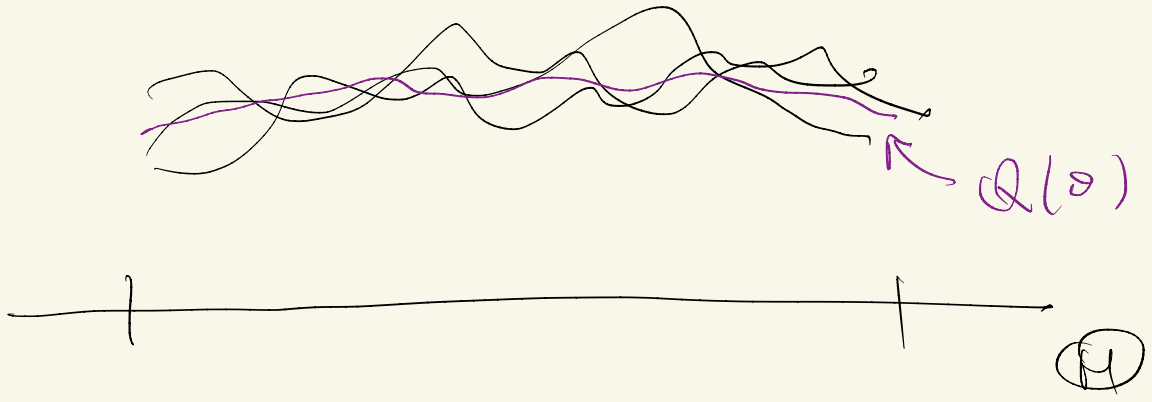
Problem

$$\sup_{\theta \in \Theta} \left| \underbrace{\frac{1}{n} \sum_{i=1}^{n} q(w_i, \theta)}_{Q_n(\theta)} - \underbrace{\mathbb{E}[q(w, \theta)]}_{Q(\theta)} \right| = o_p(1)$$

$(Q_n(\theta) : \theta \in \Theta)$ stochastic process

$(Q(\theta) : \theta \in \Theta)$ function.

$$\| Q_n - Q \|_\infty = \sup_{\theta \in \Theta} | Q_n(\theta) - Q(\theta) | = o_p(1)$$



- $\Theta \subseteq \bigcup_{\ell=1}^{J_\varepsilon} \Theta_{\ell,\varepsilon}$; $\Theta_{\ell,\varepsilon} = \{\theta \in \Theta : \|\theta - \theta_\ell\| \leq \delta\}$

- $$\mathbb{P}\left(\sup_{\theta \in \Theta} |Q_n(\theta) - Q(\theta)| > \eta \right)$$

$$\leq \mathbb{P}\left[\max_{1 \leq j \leq J_\varepsilon} \sup_{\theta \in \Theta_{\ell,\varepsilon}} |Q_n(\theta) - Q(\theta)| > \eta \right]$$

